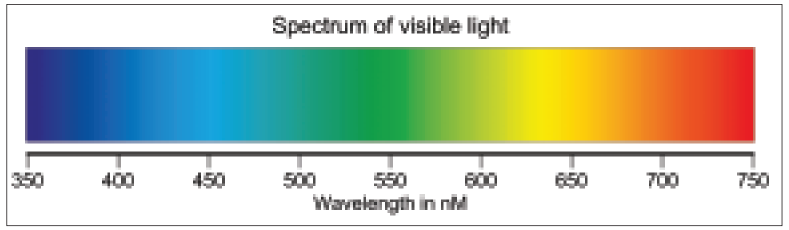
Colour Does Not Exist

Strictly speaking colour is a product of the human visual system. All that can physically be measured are the wavelength and intensity of light and not its colour. Certain mixtures of wavelengths can only be referred to as purple or brown because of a general agreement that that is what we call the visual sensations they produce. Since colour is a subjective sensation, colours are not experienced in the same way by everyone. You might think this rather difficult to prove, since we cannot directly experience the thoughts and sensations of others, but about two per cent of females and eight per cent of males have some degree of colour blindness. The most common form of colour blindness is rather misleadingly referred to as red-green colour blindness.

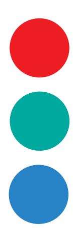
A great deal of the effort in colour science has gone in to methods of relating physical measurement to the human perception of colour. Unfortunately this is where most of the confusion over colour control and management arises. For one thing our visual perception system is not linear, so colour systems based directly on linear measurements (like the CIE XYZ 1931) do not map well to our experience of colour. For example there is quite a large portion of the green area of the CIE XYZ space that is perceived by humans as being all the same colour.

Comparing Senses – Hearing To Sight

There have been attempts to draw parallels between music and colour and to produce a system to organise colours, in the same way that musical theory provides structures such as scales and harmony for musical composition. However the human auditory and visual systems are rather different. The average human can hear a range of over 9 octaves and the common musical instrument with the widest range, the concert grand piano has a range of 8 octaves. Most human voices can cover 2 to 3 octaves. Each musical octave is a doubling of frequency. If you apply the same idea to the range of frequency of visible light you will see that there is a span of slightly less than one "octave". Visible wavelengths range from the short 380 nm (nanometre) for violet to 750 nm for the long wavelength red. Generally we speak of only seven pure colours in the visible spectrum – violet, indigo, blue, green, yellow orange and red. However there has been some success with ideas of colour harmony and various colour wheels are used by artists and designers to choose colours that go well together. Colours used for a particular design project are often organised in a palette. Colour palettes are commonly found in design software such as CorelDRAW.



Visible Spectrum



Colour Vision

Unlike our hearing, which can sense and identify sounds of a single wavelength, our eyes have three sets of sensors, known as cone cells, each of which senses light over a range of frequencies. The maximum sensitivity of one of these sets of sensors peaks in the blue, one in the green and one in the red. Outside their peak the sensitivity of each group of sensors falls off smoothly and the green sensors overlap with the blue and the red. Colours that fall outside the peak sensitivity of any of the sensors are recognised because of the balance of sensations they produce between two sensors. For example yellow stimulates both the green and the red sensors. Part of the reason our colour vision works so apparently well with just three sensors is because of the fairly limited human visual range. Our eyes also have rod cells, which respond only to light intensity.

Colour displays and printers rely on the way our eyes work and create a wide range of colours by mixing only three colours together.

Emitted, Reflected – Additive, Subtractive

Humans experience colour in two ways: by emitted light directly entering the eye or, by light reflecting from the objects around us. Mostly we experience colour through reflection and this means that the appearance of colour depends on the the nature of the surface of the illuminated object and on the spectral content of the light that is illuminating it. For example a sheet of "white" paper appears red when red light is shone on it and green when illuminated with green light. A sheet of red paper illuminated with green light appears black. This means that when working with colour and colour management the nature of the light that was and/or will be used as the illuminant must be taken into account.

Primary And Secondary - RGB Or CMY / CMYK

All the colour imaging devices we use that emit light – computer monitors, televisions and projectors use a mixture of red, green and blue light to produce a wide range of perceived colours. These devices use what's called additive colour mixing. Red, green and blue are referred to as primary colours because they relate directly to the three groups of sensors in our eyes. On the other hand, printers use cyan, magenta and yellow and so-called subtractive colour mixing. These three colours are referred to as secondary or complementary colours because cyan is white minus all the red frequencies, magenta is white minus all the green frequencies and yellow is white minus all the Blue frequencies. Or to put it another way, the three secondaries can be obtained by mixing pairs of primaries as follows; mixing blue and green light produces cyan, mixing red and blue gives you magenta and mixing red and green makes yellow.

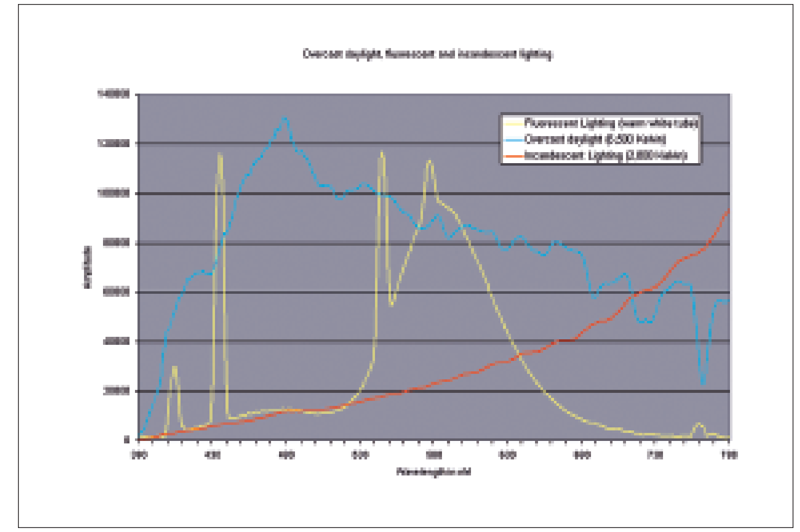
Why Are There Two Methods Of Producing Colour?

The reason that we use two methods of producing colour is this: for displays, the base state is black, so emitted light has to be added to that base state to produce colours; for printers the base state is the "white" of the paper and colours have to be selectively subtracted from that base state, leaving the required colour reflected from the page. This is why the paper used has such an effect on the final image quality and why using the more expensive, "whiter" papers results in a better image (unlike many paints, most printer inks are transparent to some degree). The expensive photo papers also have smoother surfaces and may have a gloss surface to reduce diffusion of the light striking the surface, resulting in greater image contrast. The variation in paper "white-ness" and the other variations in the nature of paper are the reason why printer drivers and colour managed applications have a choice in their output settings for paper type.

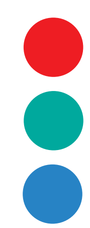
The pigments and dyes used for printing inks and the process of printing on paper aren't as successful as the three colour system used for displays. Printing cyan, magenta and yellow inks to get black usually results in a dark brown, so all ink colour printing uses an additional pigment – black - to form the CMYK system of colour printing (the letter K from the end of the word black is used to avoid confusion with blue). Photo printers extend the range of printable colours still further by adding more ink colours. Usually these are lighter versions of cyan, magenta and yellow.

White And Colour Temperature

In abstract, pure white might be regarded as emitted or reflected light that contains equal amplitudes of all the visible frequencies. Such pure white light never occurs in nature or in colour management. Our main source of illumination, sunlight, does not have a smooth or equal energy distribution and its frequency content varies depending on the time of day and on weather conditions. Artificial light is even less "pure"; for example, fluorescent light has a very uneven energy distribution with a number of large amplitude spikes at certain frequencies.



Spectra from different light sources



In practice white is always variable and relative and the human visual system is extremely good at adjusting, so that, under a wide range of lighting conditions, the areas of view that emit or reflect the widest range of frequencies with similar strength are usually seen as "white". Colour film cameras, lacking any built in compensation, don't do this and the best that can be done is to use film optimised for different lighting conditions such as "daylight" or "tungsten" film and/or to supplement these with coloured filters fitted to the lens. Perhaps one of the biggest advantages of digital cameras is that they have either, or both, auto and manual white balance. Even if the colour balance out of the camera does not look right it's easily tweaked with photo editing software, which frequently has an auto-colour or auto-white balance control.

The variations in what may be considered white mean that this has to be taken account of in colour management. Colour temperature is often used as a reference for white. It's based on the idea that a perfect black body radiates light according to its temperature. At zero Kelvin a black body radiates nothing, at 5,000 K it radiates yellow-white light rather like morning daylight, at 6,500 K it radiates blueish-white similar to overcast daylight at noon and at 9,300 K it radiates a hard blue-white light. D50 and D65 (for Daylight 5000 and 6500) are commonly used as lighting references. 9,300 K was frequently used as a default monitor setting because monitors were most efficient at this setting and produced high brightness.